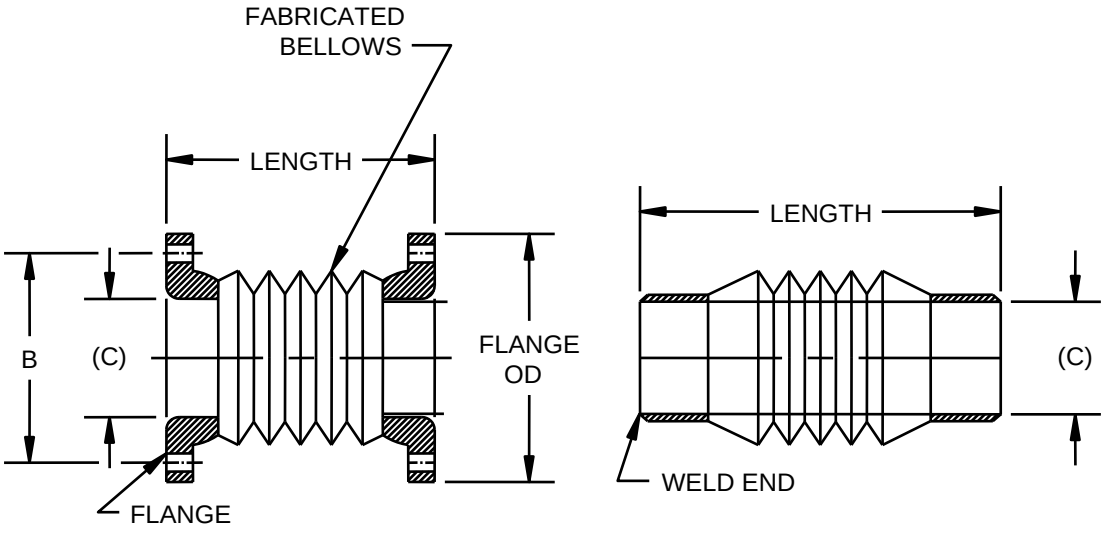
DETAIL SPECIFICATION SHEET

EXPANSION JOINTS, PIPE, METALLIC, SINGLE BELLOWS, FABRICATED

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17813.



DN	C-NPS inch (mm)	Flange OD inch (mm) 1/	B (bolt circle diameter) inch (mm) 1/
50	2 (50.8)	6 (152.4)	4.75 (120.7)
65	2.5 (76.2)	7 (177.8)	5.5 (139.7)
80	3 (76.2)	7.5 (190.5)	6 (152.4)
90	3.5 (101.6)	8.5 (215.9)	7 (177.8)
100	4 (101.6)	9 (228.6)	7.5 (190.5)
125	5 (127)	10 (254.0)	8.5 (215.9)
150	6 (152.4)	11 (279.4)	9.5 (241.3)
200	8 (203.2)	13.5 (342.9)	11.75 (298.5)
250	10 (254)	16 (406.4)	14.25 (362.0)
300	12 (304.8)	19 (482.6)	17 (431.8)
350	14 (355.6)	21 (533.4)	18.75 (476.3)
400	16 (406.4)	23.5 (596.9)	21.25 (539.8)
450	18 (457.2)	25 (635)	22.75 (577.9)
500	20 (508)	27.5 (698.5)	25 (635)
550	22 (558.8)	-	-
600	24 (609.6)	32 (812.8)	29.5 (749.3)

1/ See ASME B16.5, flange dimensions,150 pound.

FIGURE 1. Expansion joint, single bellows, fabricated.

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NOTES:

1. Dimensions are in inches. Metric equivalents are given for information only.
2. The illustration is for identification purposes only and is not intended to restrict designs and shapes of fastening devices, which conform to the requirements and are within the envelope dimensions.

FIGURE 1. Expansion joint, single bellows, fabricated – Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Construction: Fabricated bellows expansion joints shall be constructed of single-formed diaphragms joined together with circumferential seams at the inner and outer perimeters of the diaphragms, or constructed of single formed corrugations joined together with circumferential seams of the inner perimeters of the corrugations.

The classification of the expansion joints covered under this specification sheet are as follows:

Type III - shall have fabricated bellows.

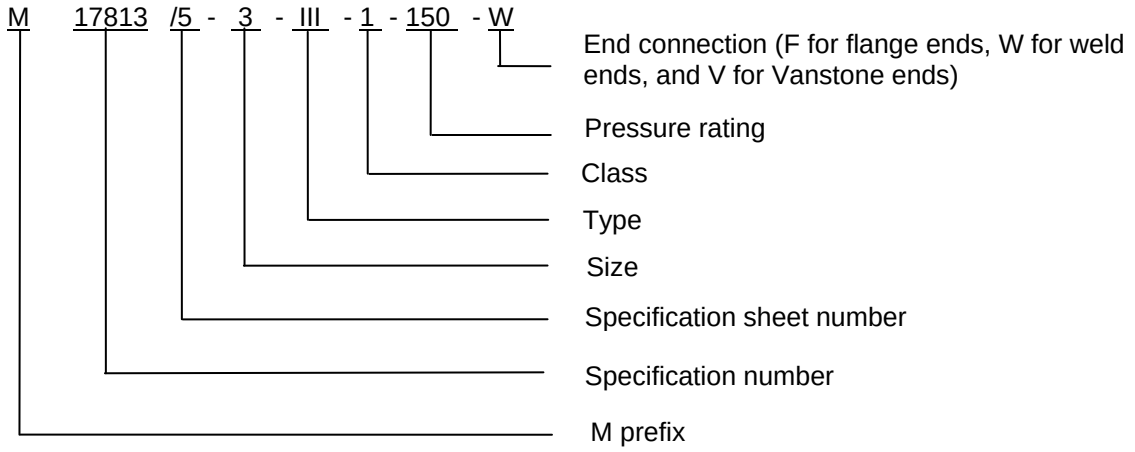
Class 1 - shall have single bellows.

Type III expansion joints shall have fabricated bellows of disk or diaphragm construction.

Materials and finishes.

Materials and finishes for the bellows and flanges shall be in accordance with MIL-DTL-17813.

The Part or Identifying Number (PIN) consists of the letters M, the basic specification number, specification sheet number, size, type, class, pressure rating, and end connection.



PIN example: M17813/5-3-III-1-150-W

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Referenced documents. In addition to MIL-DTL-17813, this document references the following:

ASME B16.5

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2009-103)

Review activities:
Army - CE
Navy – MC, SA
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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